

Total No. of Questions : 8]

SEAT No. :

PC2669

[Total No. of Pages : 2

[6354]-834

B.E. (Mechanical/Automobile)

HONORS IN MODELLING & SIMULATION OF EHV

(2019 Pattern) (Semester - VII) (302034 MJ)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Draw the neat sketch wherever necessary.

- Q1)** a) Explain A Motor Control Unit in electric vehicle with its functional features? **[4]**
- b) Explain Electronic Control Unit of electric vehicle with its types and working principal? **[8]**
- c) What is Battery/Cell Control System in EV? Explain with its functions and working principal? **[8]**

OR

- Q2)** a) Explain Torque and speed coupling in electric vehicle? **[4]**
- b) Explain EV and EHV configuration based on power electronics? **[8]**
- c) Explain the importance of Sensor Management and Integration for electric vehicle with functional and application point of view? **[8]**

- Q3)** a) Explain EV configuration and list out the components involved in it? **[8]**
- b) Explain the important feature of following (any four). **[8]**
- i) Unicycle
 - ii) Bicycle
 - iii) Dicycle
 - iv) Tricycle
 - v) Qudracycle

OR

P.T.O.

- Q4) a)** Explain front and rear wheel drive for electric vehicle with its advantages and disadvantages? [8]
- b)** Explain Propulsion and Power distribution system in electric vehicle with its main components? [8]

- Q5) a)** Explain Frame building Problems? Also list out Types of Frame Damage & How to Spot it? [8]
- b)** What do you mean by the term “DFMEA”? Explain the objective feature that can achieve the in the process of system or equipment design? [8]

OR

- Q6) a)** Explain Chassis frame layout with suitable sketch? Also list out various types of Loads on the Chassis frame? [8]
- b)** Explain the following parameters related to vehicle dynamics in detail. [8]
- i) Aerodynamic drag
 - ii) Aerodynamic lift
 - iii) Side force
 - iv) Rolling movement

- Q7) a)** Explain Durability along with Factors Affecting Electric Car Durability? [10]
- b)** Explain Fatigue analysis. List out The factors which affect the fatigue behavior? [8]

OR

- Q8) a)** Explain the phases involved in Crashworthiness Design along with its important features? [10]
- b)** What do you mean by Topology and Topography Optimization, explain with suitable example? [8]

